ATLAS: who builds what, and when

How to read it

1. One page per phase, one board per page: a column per lane, a row per step. Work down your own column; each cell is one chunk of work, the size of “Implement hyperbolic terrain generation”.
2. Everything in one row can be worked on at the same time. A chunk sits in the first row where it can start.
3. A1 means lane A, first chunk of that phase (the numbers start again in every phase). “needs” lists the chunks it waits for, your own lane’s previous chunk included; “gate” names the gate that checks it.

Four lanes, one person each

A — Platform, Look and Flat Spaces
Chunks per phase: 4, 5, 2, 2

B — Kernel and Curved Worlds
Chunks per phase: 4, 4, 2, 1

C — Scan, Room and Safety
Chunks per phase: 4, 4, 2, 3

D — Link, Station and Forge
Chunks per phase: 4, 3, 3, 2

The shared start of Phase 1

Before the lanes split. These have no dependency, so they are done in parallel.

A	Build seat, signing and Unity on every machine (lane A leads)
D	Station ready: .NET 8, Python, CUDA
C	Phones in Developer Mode and the team’s own Wi-Fi router
A B C D	Four sets of contract pages, one per lane, written from the plan

A **Then:** the Unity project and repository skeleton (lane A), then gate **G0**: contracts frozen, stubs compile.

The four phases

Phase	Chunks	Gates	Ends at
1	17	G0 to G7	G7 Phase-1 exit
2	16	G8 to G11	G11 Phase-2 exit
3	9	G12, G13	G13 FieldFit
4	8	G14, G15	G15 Golden build and submission

Phase 1

17 chunks, 6 steps, ends at gate G7 (Phase-1 exit)

Everything in one row can be worked on at the same time. A chunk can start once the chunks it needs have delivered what it waits for.

Step	A Platform, Look and Flat Spaces	B Kernel and Curved Worlds	C Scan, Room and Safety	D Link, Station and Forge
1	**All lanes The start: set-up and contracts** needs: nothing gate: **G0** Build seat, signing and Unity on every machine (A leads); Station ready: .NET 8, Python, CUDA (D); phones in Developer Mode and the team's own Wi-Fi router (C); four sets of contract pages, one per lane. Then the Unity project and repository skeleton (A), then the gate.			
2	**A1 Get the app running in stereo on both phones** needs: nothing gate: **G1**	**B1 Implement the geometry kernel and the curved-walk rig** needs: nothing gate: **G2**	**C2 Implement room scanning and the RoomModel** needs: nothing gate: none directly	**D1 Build the phone-to-Station link** needs: A1* gate: **G1** **D4 Build the Forge: object photos to furniture meshes** needs: C3* gate: **G6**
3	**A2 Build the shared look shader and the SpaceLab host** needs: A1 gate: none directly	**B2 Implement hyperbolic terrain generation** needs: A2*, B1 gate: **G3**, **G6**	**C1 Bring up AR tracking, input and the pose pipeline** needs: A1, A2* gate: **G1**, **G2**, **G5**	**D2 Build live tuning, the operator Console and telemetry** needs: A1, C1*, C3*, D1 gate: **G1**, **G5**, **G6**
4	**A3 Implement the flat and 4D spaces** needs: A2, C3* gate: **G3**, **G7**		**C3 Rebuild the room as Home and add Bump Guard** needs: A1, A2, C2 gate: **G5**, **G6**	**D3 Build the Spectator mirror and the flat-twin map** needs: A2, B2, D2 gate: **G6**
5		**B3 Implement Small Planet and Aligned Cells** needs: B2, C3 gate: **G3**, **G7**	**C4 Build onboarding and the pre-scan path** needs: C3, D1 gate: **G5**	
6	**A4 Hold the frame rate: thermal ladder, performance runs and soak** needs: A3, B3, B4*, C3, D2 gate: **G4**	**B4 Put the scanned room into the curved world** needs: B3, C2, C3 gate: **G7**		

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Gates in this phase, and what must be true to pass them

G0 Contracts frozen: every contract of §6.4 is written and stubbed; each lane's sandbox runs against the stubs

G1 First light: empty stereo scene at 60 fps on the phone; telemetry visible on the Station; anyone can install a build

G2 Curved-walk spike: five-corner loop closes within 0.25 m and three walkers from the team are comfortable at the loop-fit curvature (with a spotter: Bump Guard does not exist yet). Decides the default mode and the comfort floor

G3 Spaces in the Editor: S1–S6 run in SpaceLab on the three sample rooms; all mathematics tests green

G4 Spaces on the phone at budget: every space S0–S6 meets the Must column cold, with the room's content, on the budget_room fixture; the 15-minute soak passes

G5 Scan, Home and Bump Guard: both scan paths end in a registered Home; Bump Guard passes its scripted test in the headset

G6 Station link, four functions: mirror, tuning and Console, telemetry, and the scan and asset round-trip all pass §7.6

G7 Phase-1 exit: a first-time walker goes from self-scan to every space, watched and tuned from the Station

Phase 2

16 chunks, 4 steps, ends at gate G11 (Phase-2 exit)

Everything in one row can be worked on at the same time. A chunk can start once the chunks it needs have delivered what it waits for.

Step	A Platform, Look and Flat Spaces	B Kernel and Curved Worlds	C Scan, Room and Safety	D Link, Station and Forge
1	**A1 Build the Slicer toybox exhibit** needs: nothing gate: none directly	**B1 Generate the airway wing in hyperbolic space** needs: D2* gate: none directly	**C1 Build the Geometry Playground shell and guided walk** needs: nothing gate: none directly **C3 Build the Glued and Impossible Rooms exhibits** needs: nothing gate: none directly	**D1 Define the case contract and the illustrative case** needs: nothing gate: none directly **D3 Build the Lung panels, plan card and event logging** needs: nothing gate: **G10**
2	**A2 Wire the equation sources and finish the Playground look** needs: A1, C2*, C3 gate: none directly **A3 Implement the breathing-tumour 4D views** needs: A1, D1, D2* gate: none directly		**C2 Build captions, equation cards, action twins and the exhibit menu** needs: A1, C1 gate: none directly	**D2 Build the real-case pipeline** needs: D1 gate: none directly
3	**A4 Build the breath lane, gating and scanner tilt** needs: A3 gate: none directly	**B3 Build airway navigation: beacon, endoluminal window and Two Compasses** needs: B1, D2 gate: **G10** **B4 Build the needle sky** needs: B1, D2 gate: **G10**	**C4 Check Bump Guard in every space and run the Playground** needs: A2, C3 gate: **G8**	
4	**A5 Build the Lung tasks and director, and pass the 4D soak** needs: A4, D2 gate: **G9**, **G10**	**B2 Implement walking the airway: valve chambers and mode switch** needs: A4, B1 gate: **G10**		

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Gates in this phase, and what must be true to pass them

G8 Playground run: a first-time walker completes the guided run in two different rooms

G9 Lung 4D: the four 4D pieces run on both case packages at budget

G10 Lung airway and sky: four forks walked without rescue; sky and plan card correct

G11 Phase-2 exit: Playground and Lung Mission in one session by an outsider, hot, at budget

Phase 3

9 chunks, 3 steps, ends at gate G13 (FieldFit)

Everything in one row can be worked on at the same time. A chunk can start once the chunks it needs have delivered what it waits for.

Step	A Platform, Look and Flat Spaces	B Kernel and Curved Worlds	C Scan, Room and Safety	D Link, Station and Forge
1	**A1 Implement the two-sheet house: sheets, door and flat-twin data** *The sheet state machine* needs: nothing gate: none directly	**B1 Implement the FieldFit lens** *The lens mathematics* needs: D2* gate: none directly **B2 Build the Blind-Spot Finder** *The Blind-Spot Finder, and Scotoma River (Extended)* needs: nothing gate: none directly	**C1 Scan the second room and fit the door** *The second scan path* needs: nothing gate: none directly	**D1 Relay the peer and show the two-plan flat twin** *The peer relay (Extended)* needs: A1* gate: none directly **D2 Build the FieldFit presets and lens panel** *The field presets* needs: nothing gate: none directly
2	**A2 Load the second home and the visitor** *The second home loader* needs: A1 gate: none directly			**D3 Build the heatmap and the search task, and run the FieldFit measurement** *The Spectator heatmap and bound* needs: B1, B2, D2 gate: **G13**
3			**C2 Make the Shared House safe and shared, and walk it** *Bump Guard on the second sheet* needs: A2, C1, D1 gate: **G12**	

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Gates in this phase, and what must be true to pass them

G12 Shared House: Post Circling walked by three people with no contact

G13 FieldFit: blind spot found; search is faster with the lens than without

Phase 4

8 chunks, 3 steps, ends at gate G15 (Golden build and submission)

Everything in one row can be worked on at the same time. A chunk can start once the chunks it needs have delivered what it waits for.

Step	A Platform, Look and Flat Spaces	B Kernel and Curved Worlds	C Scan, Room and Safety	D Link, Station and Forge
1	**A1 Finish the look of every space and harden performance** *The look pass for every space and the soak on both phones* needs: nothing gate: **G14**	**B1 Prove and document the mathematics** *The maths notes* needs: nothing gate: **G15**	**C1 Polish onboarding and the pre-scan fast path** *Onboarding polished with outside walkers* needs: nothing gate: **G15**	**D1 Build the judges' dashboard and the session logs** *The judges' dashboard, and the session log with the pilot data* needs: nothing gate: none directly **D2 Produce the metrics report, the video, the deck and the offline pack** *The metrics report* needs: A1* gate: **G15**
2			**C2 Run the failure drills and the rehearsals** *The failure drills and the rehearsals with outside walkers* needs: C1, D1 gate: **G14**, **G15** **C3 Write the comfort protocol, the sign and the briefings** *The comfort protocol* needs: C1, D1 gate: **G15**	
3	**A2 Add audio, audit the comments and ship the golden build** *The audio loops and cues* needs: A1, C2 gate: **G15**			

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Gates in this phase, and what must be true to pass them

G14 Feature freeze: full path twice in a row on one phone without cooling; only fixes after this

G15 Golden build and submission: golden build runs with the Station unplugged; documents, video and metrics delivered